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Le_Hoang_Minh_Quan_Proposal

Course 66

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IARCO 2026

Research Proposal

on

Evaluating Frozen FlyVis Representations for Letter Recognition under Limited Labels and Visual Corruption

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Research topic: Computational Neuroscience and Machine Learning

Research Problem

Does a frozen, motion-pretrained FlyVis representation improve uppercase-letter recognition with limited labels and unfamiliar fonts, compared with classifiers using the same retinal input? This study will test that question and examine performance under predefined corruptions. FlyVis models a fruit-fly visual circuit using measured connectivity and parameters optimized for visual motion [1]. Applying it to letters will test whether a representation developed for one task retains useful information for another. The practical decision is whether its downstream recognition performance justifies using a more complex encoder instead of a simpler classifier.

A connectome maps neuronal connections. Here, a frozen encoder retains all its existing parameters, while a readout learns to map its responses to the 26 uppercase Latin letters. Sample efficiency will mean accuracy across specified downstream training-label budgets; robustness will mean accuracy under specified visual changes. These definitions exclude the original cost of pretraining and do not imply biological reading. A negative result would still identify a boundary of representation transfer. Conversely, beating a weak baseline would not establish practical value if a conventional neural classifier performed better at lower cost.

Letters provide a compact diagnostic task with a fixed label set and visible variation in shape. Holding out font groups will test whether a classifier recognizes a class across new visual styles rather than merely matching familiar rendered images. Controlled perturbations will then separate performance on clean examples from sensitivity to particular input changes. This restricted setting makes errors and comparisons easier to inspect. It will not establish a deployable document-recognition system, but can identify whether the representation warrants a broader transfer study.

Existing Literature

Whole-brain reconstruction provides structural information about the adult fly nervous system [2], but it is distinct from FlyVis's released visual-circuit configuration. In this study, the encoder configuration will remain fixed to specifically test whether features learned for motion processing can support a different recognition task without further adaptation. Fixing the encoder isolates the contribution of its pretrained representations, ensuring that any downstream performance reflects transferability rather than retraining effects. The primary uncertainty, therefore, concerns the utility of these pretrained responses outside their original motion task, rather than whether connectomic structure alone accounts for neural computation. I will use a lightweight readout to test this question without retraining the whole circuit on letter recognition.

FlyGM, a locomotion preprint using connectomic graphs, illustrates why structural comparisons require explicit graph controls [3]. Its results do not transfer to letter recognition. Here, an untrained FlyVis instance with identical architecture, feature pooling, and readout will test the contribution of optimized parameters. It will not isolate the causal value of biological connectivity. Such an inference would require an alternative graph with comparable pretraining, which is outside the primary study.

Google Fonts provides font files and license metadata for controlled glyph generation and lineage grouping [4]. For an optional handwriting-domain test, EMNIST ByClass preserves case distinctions, whereas EMNIST Letters merges uppercase and lowercase [5]. MNIST-C provides precedent for evaluating fixed corruption suites [6], although it does not justify the severities chosen here. Together, these resources support a leakage-aware test across held-out font lineages, controlled corruptions, and a separate handwritten domain. This study contributes a controlled comparison of a motion-pretrained FlyVis representation with retinal and untrained

controls under limited-label conditions; it does not claim novelty for transfer learning itself or claim to be the first letter-transfer study.

Research Question and Hypotheses

The central question is whether a frozen, motion-pretrained FlyVis representation improves uppercase-letter recognition under limited labels and unfamiliar fonts compared with classifiers using the same retinal input. Three hypotheses operationalize this question.

H1 predicts higher clean unseen-font accuracy, summarized across label budgets, for pretrained FlyVis than for a retinal linear classifier. This primary comparison keeps the readout family fixed while changing the representation.

H2 predicts higher clean accuracy for pretrained FlyVis than the untrained FlyVis control at 20 labels per class.

H3 predicts higher mean corrupted-test accuracy for pretrained FlyVis than retinal linear at 20 labels per class.

H2 and H3 will be secondary analyses. A multilayer perceptron (MLP) and a small convolutional neural network (CNN) will test whether any benefit remains competitive with conventional learned features. No hypothesis concerns learning stroke components and composing letters.

Methodology

A. Study Design and Dataset

The main dataset will use 70 independent font-lineage groups: 40 for training, ten for validation, and 20 for testing. Six additional groups will be reserved for pilot checks. One upright regular font per group will supply A–Z. Before assigning fonts to dataset splits, I will check each font for missing glyphs, verify its license, and record its file hash and rendering settings. Metadata and alphabet-image comparisons will identify aliases and derivatives; near-duplicate candidates will be reviewed and grouped before splitting. A seed-2026 manifest will assign eligible groups to splits. A held-out group will share no identified lineage with training or validation groups. Every transformed image will inherit its source split.

Budgets will be 5, 10, 20, and 40 training glyphs per class, sampled as nested font-group subsets with seeds 11, 22, 33, 44, and 55. Every arm will use identical subsets. Each group contributes one glyph per letter; transformations do not create additional independent labels. The 260 clean validation labels will be reported as a separate, fixed model-selection cost. Increasing a budget increases both labels and font diversity so that conclusions will concern this acquisition protocol, not a pure effect of label count. Training and model selection will use clean inputs only; test fonts and corruptions will not guide tuning.

B. Stimulus Encoding and Feature Extraction

Glyphs will be antialiased, aspect-preserving, and centered on a 512 × 512 canvas, with background luminance 0.5, black foreground, and longest bounding-box side 156 pixels. Each will move horizontally from −39 to +39 pixels across 80 frames, with a simulation step of 0.01 seconds. The class-independent trajectory provides dynamic input without encoding motion labels. The custom-stimulus interface [7] and BoxEye mean renderer [8], using extent 15 and kernel 13, will convert tensors from batch × time × height × width to batch × time × 1 × 721. Pilot inspection will check retinal coverage and unintended resizing.

The pretrained encoder will use the tutorial's flow/0000/000 model and its best optical-flow checkpoint, chosen independently of letter performance. I will lock the software revision, configuration, and checkpoint hash. I will initialize a fresh instance of the same configuration using random seed 2026, without checkpoint recovery [9], while keeping the fixed biological

constraints unchanged. Its parameter hash must differ from the pretrained model. Both encoders will remain frozen. Each clip will start from a fresh copy of a gray adaptation state computed over 0.5 seconds using the simulation API [10], preventing state transfer between examples while retaining a common onset transient.
Features will include every non-input cell type, identified from node-role metadata [11]. Responses will be averaged within a 5 × 5 grid of equal-width axial-coordinate bins spanning −15.5 to 15.5, and four temporal bins of 20 frames. Concatenation will produce 100 × K features for K retained cell types, with empty bins set to zero. The same pooling will serve both encoders. Only the selected training subset will determine feature means and standard deviations; constant features will use unit scale.

C. Comparison Models and Training

Table 1 defines all arms. They receive identical retinal samples, but their capacities differ, as shown below.

Table 1. Model arms and comparison roles

Pretrained FlyVis	Frozen pooled responses; linear readout	Main transfer system
Untrained FlyVis	Same graph, pooling and readout	Optimized-parameter control
Retinal linear	Four time-bin means × 721 receptors	Same learner, no encoder
Retinal MLP	Same retinal means; 128-unit ReLU layer	Nonlinear baseline
Small CNN	Shared framewise convolutions; temporal pooling	Learned spatial baseline

The MLP will map 2,884 inputs through 128 hidden units to 26 outputs. The CNN will map each retinal frame to a 31 × 31 axial grid with a missing-position mask, apply two padded 3 × 3 convolutions with 16 then 32 channels and ReLU, average-pool to 4 × 4 spatial bins, average within four temporal bins, and classify the concatenated features.
Retinal linear and MLP inputs will follow the same training-only standardization rule. The CNN will receive unstandardized luminance and the binary mask. Model capacity differences are intentional: the linear comparison examines accessibility to one learner family, while the neural baselines ask whether a more flexible learner can perform better. None controls the original pretraining budget. Full learning curves, not a selected favorable budget, will therefore determine the account of relative performance.
Linear readouts will use multinomial logistic regression with L2 regularization, C values 0.01, 0.1, 1, and 10, and LBFGS with a 1,000-iteration cap and tolerance 0.0001. Clean validation accuracy will select C; ties will favor stronger regularization.
Both neural baselines will use cross-entropy and Adam with a learning rate of 0.001, batches of 32, and at most 100 epochs. Early stopping will use validation loss with a patience of ten. Weight decays of 0, 0.00001, 0.0001, and 0.001 will provide four candidates per arm, budget, and seed. Selection will maximize clean validation accuracy, breaking ties by lower loss, then stronger regularization. Network initialization will use the subset seed. I will use no data augmentation during primary training. I will report parameter counts, selection trials, training time, and inference time; I will not describe equal input information as parameter or pretraining matching.

D. Evaluation Conditions and Outcomes

I will generate six corrupted conditions independently: rotations of ±10° or ±20° with seeded signs; Gaussian blur with standard deviation 3.25 or 6.5 pixels; and square occlusion with a side length of 20% or 35% of the glyph's longest dimension. Each change will precede motion generation; the gray occluder will remain attached to the glyph. I will sample its center within the original bounding box. Arms will share transformations and seeds, with pilot and test randomness separated. Actual foreground removal will be logged. These severities are design choices, not guarantees that every corrupted glyph remains unambiguous to a human.

The primary outcome will be normalized trapezoidal area under clean accuracy against log2 label budget. Because budgets double, the summary equals the weighted mean of accuracies at 5, 10, 20, and 40 labels, with weights 1, 2, 2, and 1 divided by six. The primary contrast will subtract retinal-linear performance from pretrained FlyVis. A paired 95% interval entirely above zero and a point estimate of at least two percentage points will support a practically relevant advantage. An interval below zero will contradict H1; one spanning zero will leave direction unresolved. Smaller positive estimates will not meet the practical threshold. Two points is a prespecified preference, not a predicted or powered effect.

At 20 labels per class, secondary analyses will compare pretrained and untrained clean accuracy and average accuracy over all six corrupted conditions. I will also report macro-F1, confusion matrices, individual corruption results, and each baseline's learning curve. Robustness loss will be clean minus corrupted accuracy in percentage points, and that difference divided by clean accuracy when nonzero. Absolute corrupted accuracy will accompany each loss to avoid rewarding a poor clean classifier. H2 or H3 will be contradicted by a paired interval entirely favoring its comparator; intervals spanning zero will be inconclusive. Secondary intervals will remain exploratory.

E. Statistical Analysis and Reproducibility

Uncertainty will be estimated using 10,000 paired bootstrap resamples of the test font groups, retaining all 26 letters and corruption variants within each sampled group. For each resample, the statistic will be calculated for each seed and averaged across seeds; percentile intervals and seed ranges will be reported separately. This interval conditions on the chosen training pool, checkpoint, and protocol. At the full training budget, deterministic linear fits are repeats, not independent evidence.

Model predictions will share sample identifiers. I will retain failures, nonconvergence, exclusions, configurations, environment versions, and timings; missing predictions will invalidate an arm's comparison until corrected consistently. Code, split manifests, prediction files, and permitted data-generation assets will be released with licenses and version information, allowing others to reproduce both favorable and unfavorable outcomes.

F. Pilot and Feasibility

An A/B/C smoke test on the six pilot groups will precede the main experiment. It will check input shapes, finite responses, reset reproducibility, checkpoint separation, pooled-feature dimensions, and retinal visibility. A retinal classifier must fit at least 95% of the tiny training set as an engineering check; FlyVis need not outperform it. Ten clips after warm-up will measure time and peak memory on the available hardware. The primary dataset will require 4,940 clips per encoder, including clean and corrupted tests. Float32 caches will require approximately $2 \times 4{,}940 \times 100 \times K \times 4$ bytes for both encoders, excluding buffers and retinal features.

If measured projections exceed available resources or a 24-GPU-hour ceiling, I will first cache pooled features and simulate clips sequentially. I will reduce frame count or canvas size only if

the pilot preserves retinal information, duration, and numerical stability. The declared fallback will use 20 training, eight validation, and 12 test groups with budgets of 5, 10, and 20 labels per class, followed by one severity per corruption if necessary. I will reduce the alphabet to A-L only as a last resort and will explicitly revise the scope and label costs. All changes will precede test inspection and apply across arms; failed engineering checks will stop the main run.
If resources remain, a static-input ablation will refit readouts on centered repeated-frame clips to test sweep dependence. A separate handwriting evaluation may use 100 examples per uppercase class from the official EMNIST ByClass test split, filtered through its label mapping and checked for orientation and polarity. It will use no handwriting-based tuning. Stroke reveal will remain outside the core experiment because changing presentation order alone would not establish compositional learning.

Research Topic and Expected Contribution

Expected Outcomes

If pretrained FlyVis improves both the learning curve and corrupted accuracy, this would support transfer of its motion-pretrained representation to this controlled letter-recognition task under limited supervision and input corruption. A gain over untrained FlyVis but a loss to the CNN would suggest that optimized FlyVis parameters contribute useful features without matching a task-trained neural baseline. A null or negative result would instead define a boundary of transfer and motivate narrower tests of when motion-derived representations help.
The study will test a precisely bounded claim about downstream representation utility. Its paired controls and locked evaluation will make positive, mixed, and negative outcomes informative. It will not infer biological reading or a causal advantage of connectomic wiring from classifier accuracy alone.

Limitations

Simplified dynamics and synthetic glyphs limit biological interpretation; the study will therefore treat results as evidence only for representation transfer, while optional handwriting and static-input tests probe limited extensions. Font-lineage auditing can reduce, but cannot prove the absence of, shared design ancestry. Fixing one checkpoint prevents favorable model selection but limits generalization across FlyVis encoders.

Timeline

Week 1 will establish the pilot and font manifest; week 2 will generate inputs and caches; weeks 3–4 will fit readouts and baselines; week 5 will evaluate corruptions and uncertainty; and week 6 will check reproduction and write the analysis, with two buffer weeks for implementation problems. No experiment is presented as completed.

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